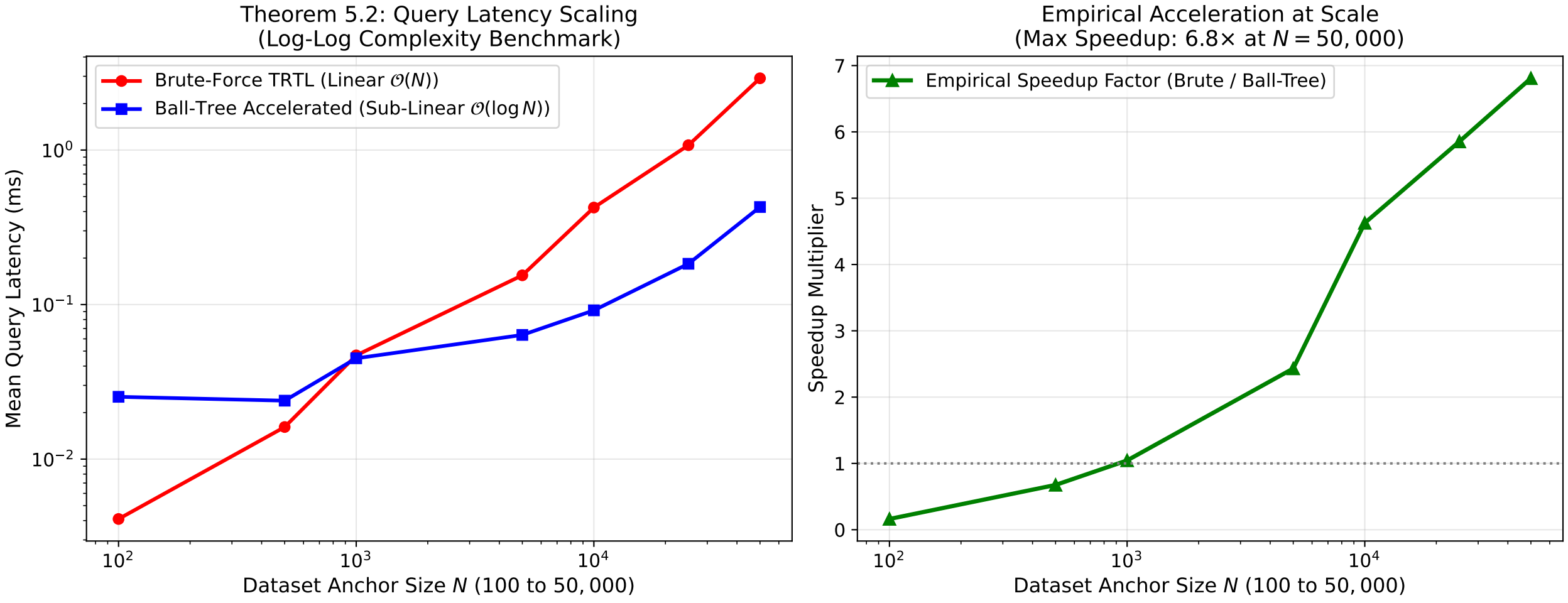
Theorem 5.2: Query Latency Scaling
(Log-Log Complexity Benchmark)
Brute-Force TRTL (Linear $\mathcal{O}(N)$)
Ball-Tree Accelerated (Sub-Linear $\mathcal{O}(\log N)$)
Mean Query Latency (ms)
10^{0}
10^{-1}
10^{-2}
10^{2}
10^{3}
10^{4}
Dataset Anchor Size N (100 to 50, 000)
Empirical Acceleration at Scale
(Max Speedup: 6.8× at $N = 50,000$)
Empirical Speedup Factor (Brute / Ball-Tree)
Speedup Multiplier
0
1
2
3
4
5
6
7
10^{2}
10^{3}
10^{4}
Dataset Anchor Size N (100 to 50, 000)